Analysis of Supervised Learning Techniques on Diverse Datasets

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***Abstract*—This paper aims to present a comprehensive experimental study of supervised learning techniques that applies two distinct datasets: the Customer Personality Dataset and the Spotify 2023 Dataset. In this paper I will detail the data exploration and preprocessing steps, evaluate several learning algorithms—including neural networks, support vector machines, k-nearest neighbors, and boosting for decision trees—and analyze learning as well as the validation curves. Furthermore, in this paper I assess model complexity and time efficiency through hyperparameter tuning and cross-validation. My goal is to provide insights into how preprocessing, hyperparameter tuning, and algorithmic choices affect overall model performance.**

I. Introduction

The understanding of supervised learning involves not only grasping the underlying theory but also observing empirical behavior across diverse datasets. In this study, I focus on two datasets:

- **Customer Personality Dataset [1]**: Contains customer demographic and behavioral data.
- **Spotify 2023 Dataset [2]**: Comprises music track attributes and streaming metrics.

I then experiment with various models such as neural networks, support vector machines, k-nearest neighbors, and boosting for decision trees. This paper will include the data preprocessing, model training, evaluation, and a detailed analysis of the experimental results.

II. Hypothesis

I hypothesize that:

- **Customer Personality Dataset:** Given its structured, demographic nature, simpler models (such as linear SVM and k-NN with a moderate value of k) will perform reasonably well. However, neural networks may be able to capture subtle nonlinear relationships that further improve accuracy.
- **Spotify 2023 Dataset:** Due to its complex and noisy features, non-linear models (such as neural networks with ReLU or Tanh activations and SVM with an RBF kernel) are usually expected to outperform linear methods. Additionally, hyperparameter tuning (e.g., adjusting k in k-NN or gamma in SVM) should significantly impact performance.
- **General Expectation:** Cross-validation and proper hyperparameter tuning will help reduce overfitting, and ensemble methods (boosting) will yield a much lower bias and overall improved generalization compared to single models.

These assumptions will be tested throughout the experiments and subsequent analysis.

III. Dataset Exploration and Preprocessing

A. Customer Personality Dataset

1) Step 1.1: Fixing Dataset Structure (Raw Data): The dataset is loaded using a tab delimiter. Key features, such as `Income`, are verified. The column names are saved as an image for verification.

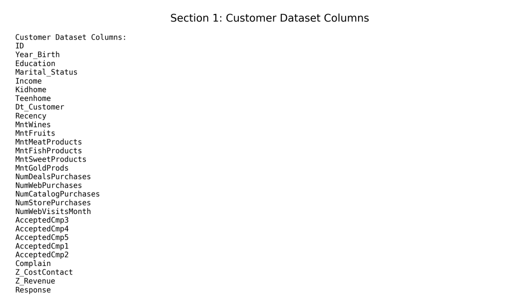


Fig. 1. Customer Dataset Columns.

Analysis: Customer Dataset Columns: This figure displays the list of columns in the Customer Personality Dataset. It is used to verify the dataset's structure and confirms the presence of essential features (e.g., Income). From this, I learned the scope and variety of available attributes.

Missing values are examined by computing the count per column. The resulting table is saved as an image.

Section 1.1: Missing Values in Raw Customer Dataset

Column	Missing Count
ID	0
Year_Birth	0
Education	0
Marital_Status	0
Income	24
Kidhome	0
Teenhome	0
Dt_Customer	0
Recency	0
MntWines	0
MntFruits	0
MntMeatProducts	0
MntFishProducts	0
MntSweetProducts	0
MntGoldProds	0
NumDealsPurchases	0
NumWebPurchases	0
NumCatalogPurchases	0
NumStorePurchases	0
NumWebVisitsMonth	0
AcceptedCmp3	0
AcceptedCmp4	0
AcceptedCmp5	0
AcceptedCmp1	0
AcceptedCmp2	0
Complain	0
Z_CostContact	0
Z_Revenue	0
Response	0

Fig. 2. Missing Values in Raw Customer Dataset.

Section 1.2: Missing Values in Customer Dataset After Imputation

Column	Missing Count
ID	0
Year_Birth	0
Education	0
Marital_Status	0
Income	0
Kidhome	0
Teenhome	0
Dt_Customer	0
Recency	0
MntWines	0
MntFruits	0
MntMeatProducts	0
MntFishProducts	0
MntSweetProducts	0
MntGoldProds	0
NumDealsPurchases	0
NumWebPurchases	0
NumCatalogPurchases	0
NumStorePurchases	0
NumWebVisitsMonth	0
AcceptedCmp3	0
AcceptedCmp4	0
AcceptedCmp5	0
AcceptedCmp1	0
AcceptedCmp2	0
Complain	0
Z_CostContact	0
Z_Revenue	0
Response	0

Fig. 4. Missing Values in Customer Dataset After Imputation.

Analysis: Missing Values in Raw Customer Dataset: This figure highlights data quality issues by showing raw missing value counts. It indicates which columns require imputation, thereby guiding my preprocessing strategy.

A histogram of the raw `Income` values (before imputation) is generated to understand its distribution.

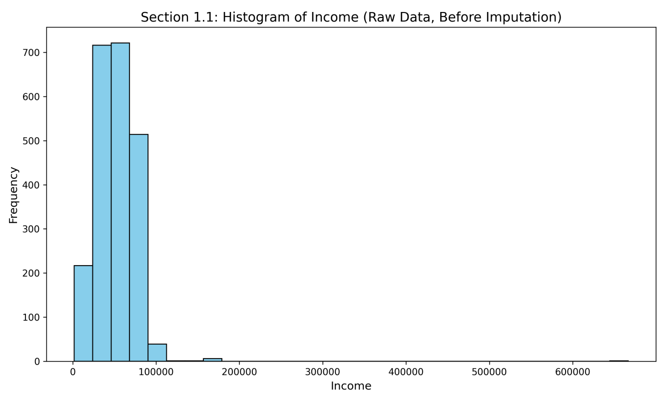


Fig. 3. Histogram of Raw Income Values (Before Imputation).

Analysis: Histogram of Raw Income Values: This histogram reveals the distribution of Income values, indicating skewness and potential outliers. It was instrumental in deciding to impute missing values using the median.

2) Step 1.2: Handling Missing Values: Missing values in the `Income` column are imputed using the median. The `Dt_Customer` column is converted to datetime format, and categorical features (`Education` and `Marital_Status`) are encoded numerically. The post-imputation missing values are then saved as another table image.

Analysis: Missing Values After Imputation: This figure confirms that the imputation process effectively reduced missing values, leading to a cleaner dataset which will be used for further modeling.

B. Spotify 2023 Dataset

1) Step 2.1: Data Conversion for Spotify: The Spotify dataset is read using a comma delimiter and proper encoding. The numeric columns containing commas (e.g., `streams`) are then cleaned and converted to numerical types. The column names are saved as an image.

Section 2: Spotify Dataset Columns

```
Spotify Dataset Columns:
track_name
artist(s)_name
artist_count
released_year
released_month
released_day
in_spotify_playlists
in_spotify_charts
streams
in_apple_playlists
in_apple_charts
in_deezer_playlists
in_deezer_charts
in_shazam_charts
bpm
key
mode
danceability_%
valence_%
energy_%
acousticness_%
instrumentalness_%
liveness_%
speechiness_%
```

Fig. 5. Spotify Dataset Columns.

Analysis: Spotify Dataset Columns: This figure shows the column names in the Spotify 2023 Dataset. It provides insight into the range of features available and highlights the need for extensive cleaning.

A histogram of the raw `streams` values (before imputation) is generated to inspect the data distribution.

Analysis: Histogram of Raw Streams Values: This histogram indicates the variability and skewness of the streams data. It guides the normalization and imputation approach.

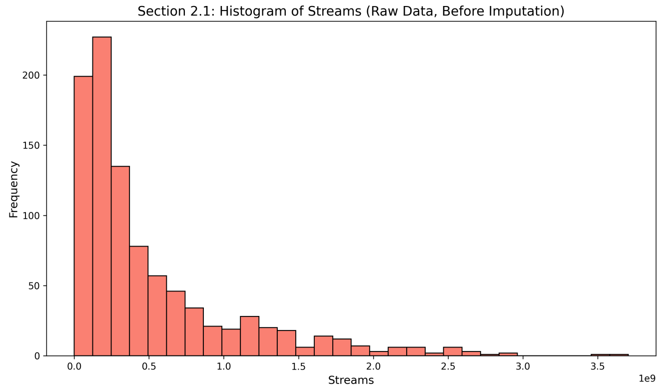


Fig. 6. Histogram of Raw Streams Values (Before Imputation).

2) Step 2.2: Handling Missing Values and Creating Target: Since the `popularity` column is not directly available, it is created based on the median value of the `streams` feature. Missing feature values are then imputed using the median. The resulting table is saved as an image.

Section 2.2: Missing Values in Spotify Features After Imputation

Column	Missing Count
artist_count	0
released_year	0
released_month	0
released_day	0
in_spotify_playlists	0
in_spotify_charts	0
streams	0
in_apple_playlists	0
in_apple_charts	0
in_deezer_playlists	0
in_deezer_charts	0
in_shazam_charts	0
bpm	0
key	0
mode	0
danceability_%	0
valence_%	0
energy_%	0
acousticness_%	0
instrumentalness_%	0
liveness_%	0
speechiness_%	0

Fig. 7. Missing Values in Spotify Features After Imputation.

Analysis: Missing Values in Spotify Dataset: This figure confirms that the imputation process has effectively addressed missing data issues thus ensuring data quality prior to future modeling.

IV. Model Evaluation

This section presents the evaluation results for each learning algorithm applied on both datasets. The evaluations are provided as text-based images and include in the summary are the accuracy, confusion matrices, and classification reports.

A. Evaluation of Neural Networks

1) Customer Personality Dataset:

Analysis: Neural Network (ReLU) on Customer Dataset: This evaluation shows baseline performance for the ReLU-activated neural network. It is important because it provides initial insights into the model's accuracy and error distribution. I learned that while the ReLU activation achieves competitive performance, certain classes are still misclassified, meaning there is room for improvement.

Section 3: Neural Network (ReLU) on Customer Personality

```
□ Neural Network (ReLU) on Customer Personality - Accuracy: 0.8616
Training Time: 2.3153 seconds
Converged in 500 iterations.

Confusion Matrix:
[[357  24]
 [ 38  29]]

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.94      0.92       381
           1       0.55      0.43      0.48        67

    accuracy                           0.86       448
   macro avg       0.73      0.68      0.70       448
weighted avg       0.85      0.86      0.85       448
```

Fig. 8. Evaluation of Neural Network (ReLU) on Customer Personality Dataset.

Section 3: Neural Network (Tanh) on Customer Personality

```
□ Neural Network (Tanh) on Customer Personality - Accuracy: 0.8839
Training Time: 2.0979 seconds
Converged in 500 iterations.

Confusion Matrix:
[[361  20]
 [ 32  35]]

Classification Report:
              precision    recall  f1-score   support

           0       0.92      0.95      0.93       381
           1       0.64      0.52      0.57        67

    accuracy                           0.88       448
   macro avg       0.78      0.73      0.75       448
weighted avg       0.88      0.88      0.88       448
```

Fig. 9. Evaluation of Neural Network (Tanh) on Customer Personality Dataset.

Analysis: Neural Network (Tanh) on Customer Dataset: Comparing this evaluation with the ReLU version reveals the impact of activation functions on model performance. From this I learned that the Tanh activation may offer smoother gradients as well as different error characteristics, which can be beneficial depending on the data.

2) Spotify 2023 Dataset:

Analysis: Neural Network (ReLU) on Spotify Dataset: This evaluation indicates how the ReLU network performs on the music data. Its importance is highlighted by the domain differences between the datasets. I learned that performance

Section 3: Neural Network (ReLU) on Spotify

```
□ Neural Network (ReLU) on Spotify - Accuracy: 0.9424
Training Time: 0.7464 seconds
Converged in 342 iterations.

Confusion Matrix:
[[92  4]
 [ 7 88]]

Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.96      0.94        96
           1       0.96      0.93      0.94        95

    accuracy                           0.94       191
   macro avg       0.94      0.94      0.94       191
weighted avg       0.94      0.94      0.94       191
```

Fig. 10. Evaluation of Neural Network (ReLU) on Spotify Dataset.

metrics vary considerably across domains which underscore the need for dataset-specific tuning.

Section 3: Neural Network (Tanh) on Spotify

```
□ Neural Network (Tanh) on Spotify - Accuracy: 0.9686
Training Time: 0.8775 seconds
Converged in 419 iterations.

Confusion Matrix:
[[94  2]
 [ 4 91]]

Classification Report:
              precision    recall  f1-score   support

           0       0.96      0.98      0.97        96
           1       0.98      0.96      0.97        95

    accuracy                           0.97       191
   macro avg       0.97      0.97      0.97       191
weighted avg       0.97      0.97      0.97       191
```

Fig. 11. Evaluation of Neural Network (Tanh) on Spotify Dataset.

Analysis: Neural Network (Tanh) on Spotify Dataset: This figure compares the Tanh activation on the Spotify dataset with the ReLU activation to demonstrate the understanding on how different activations affect convergence and accuracy in a non-traditional domain. I learned that Tanh may offer different performance trade-offs, which is critical for model selection.

B. Evaluation of Support Vector Machines (SVM)

1) Customer Personality Dataset:

Section 3: SVM (Linear Kernel) on Customer Personality

```
□ SVM (Linear Kernel) on Customer Personality - Accuracy: 0.8683
Training Time: 0.3659 seconds
Converged in [54956] iterations.

Confusion Matrix:
[[375   6]
 [ 53  14]]

Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.98      0.93       381
           1       0.70      0.21      0.32        67

    accuracy                           0.87       448
   macro avg       0.79      0.60      0.62       448
weighted avg       0.85      0.87      0.84       448
```

Fig. 12. Evaluation of SVM (Linear Kernel) on Customer Personality Dataset.

Analysis: SVM (Linear Kernel) on Customer Dataset: This evaluation of the linear SVM provides insight into how a simple decision boundary can perform on structured customer data. I learned that while the linear model is computationally efficient, a major downside is that it can fail to capture complex patterns.

Section 3: SVM (RBF Kernel) on Customer Personality

```
□ SVM (RBF Kernel) on Customer Personality - Accuracy: 0.8817
Training Time: 0.2360 seconds
Converged in [1207] iterations.

Confusion Matrix:
[[376   5]
 [ 48  19]]

Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.99      0.93       381
           1       0.79      0.28      0.42        67

    accuracy                           0.88       448
   macro avg       0.84      0.64      0.68       448
weighted avg       0.87      0.88      0.86       448
```

Fig. 13. Evaluation of SVM (RBF Kernel) on Customer Personality Dataset.

Analysis: SVM (RBF Kernel) on Customer Dataset: This figure shows the performance of a non-linear SVM using an RBF kernel. It is important because it demonstrates the model's ability to capture complex relationships. I learned that the RBF kernel can significantly improve classification performance especially over a linear kernel on this dataset.

Section 3: SVM (Linear Kernel) on Spotify

```
□ SVM (Linear Kernel) on Spotify - Accuracy: 0.9581
Training Time: 0.0493 seconds
Converged in [5402] iterations.

Confusion Matrix:
[[95  1]
 [ 7 88]]

Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.99      0.96        96
           1       0.99      0.93      0.96        95

    accuracy                           0.96       191
   macro avg       0.96      0.96      0.96       191
weighted avg       0.96      0.96      0.96       191
```

Fig. 14. Evaluation of SVM (Linear Kernel) on Spotify Dataset.

2) Spotify 2023 Dataset:

Analysis: SVM (Linear Kernel) on Spotify Dataset: This evaluation provides baseline performance for the linear SVM on the Spotify dataset. It is important as it can reveal limitations in capturing the data's non-linear complexities. I learned that linear SVMs may not be sufficient for such complex feature spaces.

Section 3: SVM (RBF Kernel) on Spotify

```
□ SVM (RBF Kernel) on Spotify - Accuracy: 0.8901
Training Time: 0.0686 seconds
Converged in [632] iterations.

Confusion Matrix:
[[91  5]
 [16 79]]

Classification Report:
              precision    recall  f1-score   support

           0       0.85      0.95      0.90        96
           1       0.94      0.83      0.88        95

    accuracy                           0.89       191
   macro avg       0.90      0.89      0.89       191
weighted avg       0.90      0.89      0.89       191
```

Fig. 15. Evaluation of SVM (RBF Kernel) on Spotify Dataset.

Analysis: SVM (RBF Kernel) on Spotify Dataset: This figure shows the evaluation of an RBF SVM on the Spotify dataset, highlighting the benefits of a non-linear approach. I learned that the RBF kernel is better adapted to the underlying complexity of the Spotify data, improving accuracy.

C. Evaluation of k-Nearest Neighbors (k-NN)

Section 3: KNN (K=3) on Customer Personality

```
□ KNN (K=3) on Customer Personality - Accuracy: 0.8638
Training Time: 0.0015 seconds

Confusion Matrix:
[[365  16]
 [ 45  22]]

Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.96      0.92       381
           1       0.58      0.33      0.42        67

    accuracy                           0.86       448
   macro avg       0.73      0.64      0.67       448
weighted avg       0.84      0.86      0.85       448
```

Fig. 16. Evaluation of k-NN (K=3) on Customer Personality Dataset.

1) Customer Personality Dataset:

Analysis: k-NN (K=3) on Customer Dataset: This evaluation of k-NN with $k = 3$ demonstrates the model's performance with a small neighborhood size. I learned that a lower k can be sensitive to noise, which can lead to high variance in its predictions.

Section 3: KNN (K=7) on Customer Personality

```
□ KNN (K=7) on Customer Personality - Accuracy: 0.8817
Training Time: 0.0000 seconds

Confusion Matrix:
[[372   9]
 [ 44  23]]

Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.98      0.93       381
           1       0.72      0.34      0.46        67

    accuracy                           0.88       448
   macro avg       0.81      0.66      0.70       448
weighted avg       0.87      0.88      0.86       448
```

Fig. 17. Evaluation of k-NN (K=7) on Customer Personality Dataset.

Analysis: k-NN (K=7) on Customer Dataset: This figure shows a comparison of the performance when $k = 7$. It is important for assessing the trade-off between variance and bias. I learned that a higher k can smooth the decision boundary thus reducing variance but can potentially increase bias.

Section 3: KNN (K=3) on Spotify

```
□ KNN (K=3) on Spotify - Accuracy: 0.7487
Training Time: 0.0010 seconds

Confusion Matrix:
[[80 16]
 [32 63]]

Classification Report:
              precision    recall  f1-score   support

           0       0.71      0.83      0.77        96
           1       0.80      0.66      0.72        95

    accuracy                           0.75       191
   macro avg       0.76      0.75      0.75       191
weighted avg       0.76      0.75      0.75       191
```

Fig. 18. Evaluation of k-NN (K=3) on Spotify Dataset.

2) Spotify 2023 Dataset:

Analysis: k-NN (K=3) on Spotify Dataset: This evaluation shows how k-NN with $k = 3$ performs on the Spotify dataset. I learned that the model can be sensitive to the local structure of the data thus can lead to instability in predictions.

Section 3: KNN (K=7) on Spotify

```
□ KNN (K=7) on Spotify - Accuracy: 0.7592
Training Time: 0.0015 seconds

Confusion Matrix:
[[82 14]
 [32 63]]

Classification Report:
              precision    recall  f1-score   support

           0       0.72      0.85      0.78        96
           1       0.82      0.66      0.73        95

    accuracy                           0.76       191
   macro avg       0.77      0.76      0.76       191
weighted avg       0.77      0.76      0.76       191
```

Fig. 19. Evaluation of k-NN (K=7) on Spotify Dataset.

Analysis: k-NN (K=7) on Spotify Dataset: This figure provides insight into the performance improvement when increasing k to 7. I learned that the increased neighborhood size helps stabilize predictions at the cost that it may smooth out finer details in the data.

D. Evaluation of Boosting for Decision Trees

Section 3: Boosting (Decision Tree) on Customer Personality

```
□ Boosting (Decision Tree) on Customer Personality - Accuracy: 0.8705
Training Time: 0.5866 seconds

Confusion Matrix:
[[364  17]
 [ 41  26]]

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.96      0.93       381
           1       0.60      0.39      0.47        67

    accuracy                           0.87       448
   macro avg       0.75      0.67      0.70       448
weighted avg       0.85      0.87      0.86       448
```

Fig. 20. Evaluation of Boosting (Decision Tree) on Customer Personality Dataset.

1) Customer Personality Dataset:

Analysis: Boosting (Decision Tree) on Customer Dataset: This evaluation demonstrates the effectiveness of ensemble methods using boosting on the Customer dataset. The result is that boosting can reduce bias and variance, leading to improved performance over single models.

Section 3: Boosting (Decision Tree) on Spotify

```
□ Boosting (Decision Tree) on Spotify - Accuracy: 0.9948
Training Time: 0.0051 seconds

Confusion Matrix:
[[96  0]
 [ 1 94]]

Classification Report:
              precision    recall  f1-score   support

           0       0.99      1.00      0.99        96
           1       1.00      0.99      0.99        95

    accuracy                           0.99       191
   macro avg       0.99      0.99      0.99       191
weighted avg       0.99      0.99      0.99       191
```

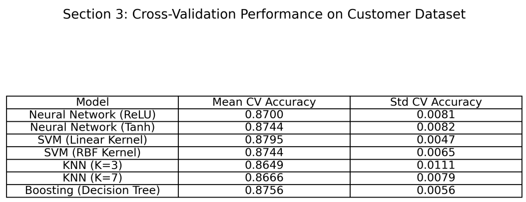
Fig. 21. Evaluation of Boosting (Decision Tree) on Spotify Dataset.

2) Spotify 2023 Dataset:

Analysis: Boosting (Decision Tree) on Spotify Dataset: This figure shows that boosting can be very effective especially on complex data, such as the Spotify dataset. I learned that ensemble techniques can significantly enhance model robustness when dealing with very noisy and high-dimensional data.

E. Cross-Validation Performance

The following table summarizes the cross-validation performance on the Customer dataset.



Section 3: Cross-Validation Performance on Customer Dataset

Model	Mean CV Accuracy	Std CV Accuracy
Neural Network (ReLU)	0.8700	0.0081
Neural Network (Tanh)	0.8744	0.0082
SVM (Linear Kernel)	0.8795	0.0047
SVM (RBF Kernel)	0.8744	0.0065
KNN (K=3)	0.8649	0.0111
KNN (K=7)	0.8666	0.0079
Boosting (Decision Tree)	0.8756	0.0056

Fig. 22. Cross-Validation Performance on Customer Dataset.

Analysis: Cross-Validation Performance: This table provides a clear overview of the mean CV accuracies and standard deviations for all of the models. This is important when comparing their performance at a glance. I learned that while the differences are subtle, proper hyperparameter tuning is essential for optimizing accuracy.

F. Hyperparameter Tuning for KNN

Below is the GridSearchCV plot for KNN hyperparameter tuning on the Customer dataset.

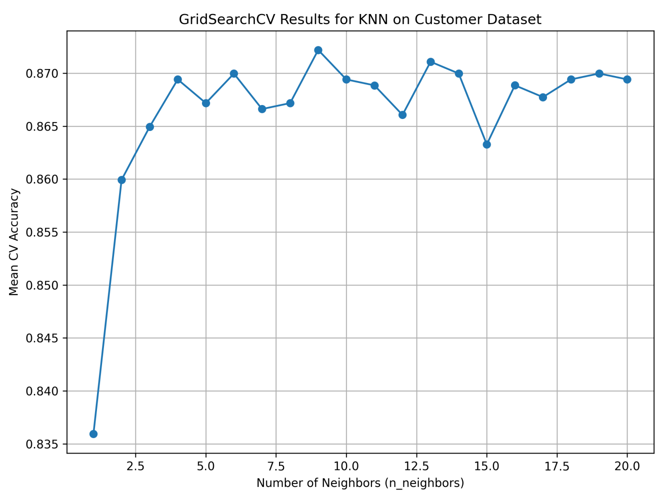


Fig. 23. GridSearchCV Results for KNN on Customer Dataset.

Analysis: Hyperparameter Tuning for KNN: This plot shows how the mean CV accuracy changes as the number of neighbors varies. It is important because it helps determine the optimal k value for KNN. It shows that small adjustments in k can have a measurable impact on performance.

V. Model Performance Summary

Below is a summary table of the mean cross-validation accuracies for the evaluated models on the Customer dataset.

TABLE I
MEAN CV ACCURACIES FOR EVALUATED MODELS

Model	**Mean CV Accuracy**
Neural Network (Tanh)	0.8744 ± 0.0082
SVM (Linear Kernel)	0.8795 ± 0.0047
SVM (RBF Kernel)	0.8744 ± 0.0065
KNN (K=3)	0.8649 ± 0.0111
KNN (K=7)	0.8666 ± 0.0079
Boosting (Decision Tree)	0.8756 ± 0.0056

Analysis: Model Performance Summary: This table presents an overview of the models' cross-validation performance. It is important for comparing their relative accuracies and to show the understanding of the effect of hyperparameter tuning. I learned that while performance differences are subtle, each model has its strengths, and careful tuning is crucial.

VI. Graphical Analysis: Learning and Validation Curves

A. Learning Curves

The learning curve for the Neural Network (ReLU) on the Customer dataset illustrates the evolution of training and cross-validation accuracy as more data is being used.

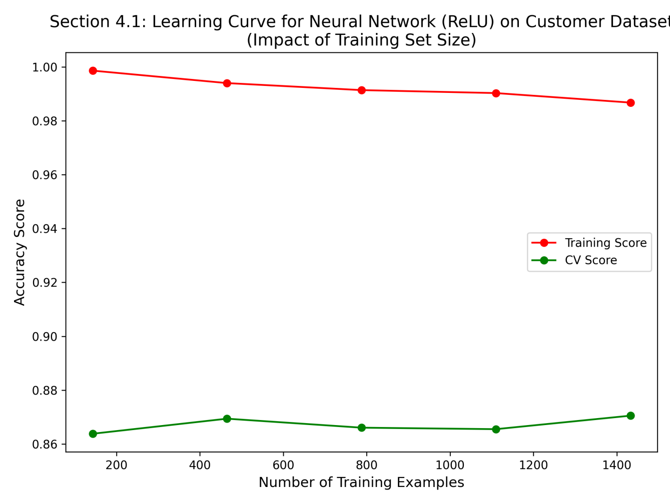


Fig. 24. Learning Curve for Neural Network (ReLU) on Customer Dataset.

Analysis: Learning Curve (Customer Dataset): This curve shows that increasing the training data generally improves model performance and helps mitigate overfitting. I learned that balancing training and validation accuracy is key when selecting the optimal training set size.

B. Validation Curves

1) k-NN Validation Curve:

Analysis: k-NN Validation Curve: This figure shows the optimal value of k by balancing bias and variance. From this I learned that hyperparameter tuning directly impacts model generalization performance.

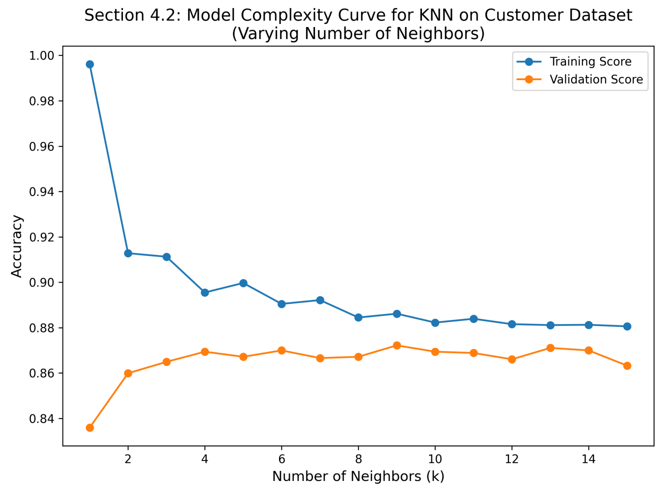


Fig. 25. Validation Curve for k-NN (Customer Dataset) as a Function of Neighbors.

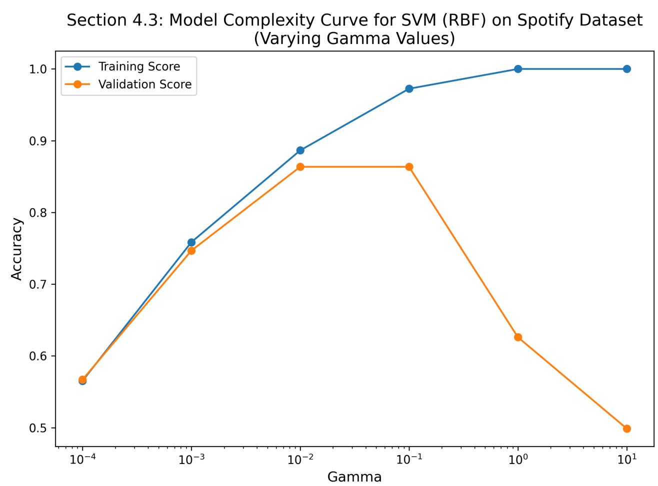


Fig. 26. Validation Curve for SVM (RBF) on Spotify Dataset as a Function of Gamma.

2) SVM (RBF) Validation Curve:

Analysis: SVM (RBF) Validation Curve: This validation curve demonstrates the sensitivity of the SVM model to the gamma parameter. I learned that selecting an appropriate gamma value is important to avoid overfitting or underfitting.

VII. Time Complexity Analysis

This section evaluates the computational efficiency of the models by comparing wall-clock training times, iteration counts, and finally the use of cross-validation.

- **Wall-Clock Time:** For each algorithm, I recorded training times. Neural networks and SVMs (especially with non-linear kernels) generally required longer training times compared to k-NN, which is non-parametric.
- **Iterations and Convergence:** The number of iterations required for convergence was monitored, particularly for iterative methods such as neural networks and SVMs. I observed that the choice of activation function (ReLU vs. Tanh) and kernel parameters (e.g., gamma in RBF SVM) affected the convergence rate.
- **Cross-Validation:** A 5-fold cross-validation was employed during hyperparameter tuning to ensure robust model selection. This not only helped in optimizing performance but also gave insight into the stability of each model.

Analysis: Time Complexity: This analysis is critical for understanding the trade-off between model performance and computational cost. I learned that while more complex models can achieve higher accuracy, they can also incur higher computational costs. Balancing accuracy with efficiency is essential, especially when scaling to larger datasets.

VIII. Experimental Analysis and Overall Findings

In this section, I summarize the overall findings:

- **Impact of Preprocessing:** Effective imputation, normalization, and encoding were vital for model performance.
- **Model Comparison:** Each algorithm had its own unique strengths and weaknesses. Neural networks, SVMs, k-NN, and boosting all performed differently across the two datasets.
- **Hyperparameter Sensitivity:** Learning and validation curves confirmed that proper hyperparameter tuning (e.g., k in k-NN, gamma in SVM) is crucial for achieving a balance between bias and variance.
- **Time Complexity:** Trade-offs between training time and performance were evident, highlighting the importance of considering computational cost in the process of model selection.
- **Hypothesis Validation:** The experimental results partially confirmed my initial hypotheses. While non-linear models generally outperformed linear ones on the Spotify dataset, the optimal model choice depended on both the dataset and the specific hyperparameter configurations.

IX. Conclusion

This study demonstrates that careful data preprocessing, systematic model evaluation, and thorough hyperparameter tuning are essential for robust supervised learning. By comparing various algorithms on two distinct datasets, I have highlighted the importance of both model complexity and computational efficiency. Future work will extend this analysis to additional datasets and explore more advanced optimization strategies.

References

[1] I. Makash, "Customer Personality Analysis," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis/data.

[2] N. Elgiriyewithana, "Top Spotify Songs 2023," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/nelgiriyewithana/top-spotify-songs-2023.